EXAM PREPARATION

1. **Heart of the heart is:**
   a) SA Node
   b) AV Node
   c) Bundle of HIS
   d) Purkinji Fiber

2. **Which of the following has the thickest wall?**
   a) Right ventricle
   b) Left ventricle
   c) Right atrium
   d) Left atrium
   Explanation: Left ventricle has the thickest wall in the human heart.

3. Pacemaker is _____
a) AV node
b) SA node
c) Bundle of HIS
d) Ventricle muscles

Explanation: Pacemaker is SA node. SA node is located in upper lateral wall of right atrium.

4. SA node is located in _____
a) upper lateral wall of right atrium
b) lower lateral wall of left atrium
c) lower lateral wall of right atrium
d) upper lateral wall of left atrium

Explanation: SA node is located in upper lateral wall of right atrium.

5. "Bundle of HIS" is made up of _____
a) nervous tissue supplied to ventricles
b) nervous tissue supplied to heart
c) muscular tissue supplied to heart
d) muscular tissue supplied to ventricles

Explanation: Bundle of HIS is made up of muscular tissue supplied to ventricles.

6. Blood enters into the heart because of muscles of _____

a) atria relax

b) ventricles contract

c) ventricles relax

d) atria contract

Explanation: Blood enters the heart through large veins called as vena cava. Blood enters the heart when the muscles of atria relax and it is pumped out when it is contracted.

7. The opening of right atrium into right ventricle is guarded by _____

a) mitral valve

b) tricuspid valve

c) bicuspid valve

d) aortic semilunar valve

Explanation: The tricuspid valve is on the right dorsal side of the mammalian heart between the right atrium and right ventricle. The function of the valve is to prevent back flow of blood into right atrium.

8. Mitral valve is present between _____

a) right atrium and left ventricle

b) right and left ventricle

c) left ventricle and aorta

d) left atrium and left ventricle

Explanation: Mitral valve is a valve that lets blood flow from one chamber of the heart, the left atrium to another called the left ventricle. In mitral valve prolapsed part of mitral valve slips backward loosely into the chamber called the left atrium.

c) left ventricle and aorta

d) left atrium and left ventricle

9.Cardiac output is determined by _____
a) heart rate
b) stroke volume
c) blood flow
d) heart rate and stroke volume

Answer: d
Explanation: Stroke volume is an amount of blood pumped by the left ventricle of the heart in one contraction. The heart rate for adults ranges from 60-100 beats per minute.

10. In human being the duration of cardiac cycle is _____
a) 0.008 sec
b) 0.5 sec
c) 0.8 sec
d) 8 sec

Answer: c
Explanation: The duration of the cardiac cycle is reciprocal of heart rate. Assuming a heart rate of 75 beats per minute, each cycle takes 0.8sec.

11. The volume of blood each ventricle pumps out during a cardiac cycle is about _____
a) 70 ml
b) 5000 ml
c) 7 L
d) 1200 ml

Answer: a
Explanation: Cardiac output is a volume of blood pumped by the heart in a minute. It is equal to the heart rate multiplied by the heart stroke, so if there are 70 beats per minute and 70ml blood is ejected with each beat of the heart, the cardiac output is 4900ml per minute

12. What is the definition of the cardiac cycle?
a) The contraction of the atria
b) Circulation of the blood in the heart
c) The contraction and relaxation of the ventricles

d) It is a sequence of event that occurs during one complete heartbeat

Answer: d
Explanation: The cardiac cycle refers to the sequence of mechanical and electrical events that repeats with every heartbeat. It includes diastole and systole.

13. Name the three phases of cardiac cycle in the order _____
a) early diastole, mid-to-late diastole, ventricular systole
b) ventricular systole, mid-to-late diastole, Early diastole
c) mid-to-late diastole, Ventricular systole, Early diastole
d) early diastole, Ventricular systole, mid-to-late diastole

Answer: c
Explanation: Every beat of the heart involves 3 major stages. They are mid-to-late diastole, Ventricular systole, Early diastole.

14. An electrocardiogram is a graphic illustration of _____
a) cardiac conduction system
b) cardiac cycle
c) cardiac output
d) systemic and pulmonary circuits

Answer: a
Explanation: Electrocardiogram is a test which measures the electrical activity of the heart. It records the heart's rhythm and activity on a moving strip of paper or a line on the screen.

15. The P wave of the ECG represents _____
a) ventricular depolarization
b) atrial depolarization
c) ventricular repolarization
d) atrial systole

Answer: b
Explanation: P wave is the first wave. It represents atrial depolarization. P wave is followed by QRS complex and T wave.

16. Which of the following is not true for ventricular systole?
a) The ventricles relax

b) The ventricles contract
c) The semi lunar valves close
d) The atrioventricular valves open

Answer: a
Explanation: Systole is the last stage of the heartbeat. It happens right after the diastole.

17. The two distinct heart sounds during the cardiac cycle are lub and dup.
a) True
b) False

Answer: a
Explanation: The two distinct heart sounds during the cardiac cycle is lub and dup. The first heart sound is lub and is followed by dup.

18. Heart beat initiates from _____
a) Purkinji fibers
b) SA node
c) Bundle of HIS
d) Auriculo ventricular node

Answer: b
Explanation: The SA node is heart's natural pacemaker. Heart beat initiates from SA node.

19. An artificial pace maker is implanted and connected to the heart in patients _____
a) suffering from arteriosclerosis
b) having high blood pressure
c) with irregularity in heart rhythm
d) having 90% blockage

Answer: c
Explanation: A pace maker is a medical device which uses electrical impulses, delivered by the electrodes contacting the heart muscles to regulate the beating of the heart.

20. A heat murmurs indicates a defective _____

a) Heart valves

b) Bundle of HIS

c) SA node

d) AV node

Answer: a

Explanation: A heart murmur can be a symptom of serious heart diseases. A heart murmur indicates defective heart valve.

21. Choose the correct pathway of the transmission of impulses in heartbeat.

a) AV node- SA node- Bundle of HIS- Purkinji fibres

b) SA node- AV node- Bundle of HIS- Purkinji fibres

c) SA node- Bundle of HIS- AV node- Purkinji fibres

d) SA node Purkinji fibres- Bundle of HIS- AV node

Answer: b

Explanation: The heart impulse starts from SA node and ends in Purkinji fibres.

22. In ECG the depolarization of atria is indicated by _____

a) P wave

b) Q wave

c) R wave

d) S wave

Answer: a

Explanation: In ECG the depolarization of atria is indicated by P wave. This happens right before the atria contracts and push blood into the ventricles.

23. Which of the following pair is correctly matched?

a) Dup- Sudden opening of semilunar valves at the beginning of ventricle diastole

b) Lubb- Sharp closing of AV valves at the beginning of ventricular systole

c) Pulsation of radial artery valves in blood vessels

d) Initiation of heart beat- Purkinji fibers

Answer: b

Explanation: The lub is the first heart sound and is caused by the closure of mitral

valve and tricuspid valves at the startup systole. The second sound dub is caused by the closer of aortic and pulmonic valve marking the end of systole.

24. Which instrument is used to listen to the internal sounds of the human body?

a) Stethoscope
b) Sphygmomanometer
c) Reflex hammer
d) Suction device

Answer: a
Explanation: Stethoscope is an acoustic medical device for auscultation. It has a small disc shaped resonator that is placed against the chest and two tubes connected to earpieces.

25. The following are the major functional physiological systems of the body except?

A. Cardiovascular system

B. Respiratory system

C. Electrocardiogram system

D. Nervous system

26. Physiological Signals are generated by the body during the functioning of various physiological systems. Therefore, physiological signals hold information which can be extracted from these signals to find out the state of functioning of these physiological systems, true or false?

A. True

B. False

27. Which one of the following is not a transducer?

A. Thermocouple

B. Photovoltaic

C. Electrooculogram

D. Moving coil generator

28. Electroencephalogram is obtained from bioelectrical signals from the?

A. Brain

B. Heart

C. Muscles

D. Retina

29. The following are resistive passive transducers, except?

A. Strain gauge

B. Potentiometer

C. Linear variable differential transformer (LVDT)

D. Photodiode

30. The capacitance of a passive capacitance transducer depends on the following factors, except?

A. Distance between the two parallel plates

B. The area of the two parallel plates

C. Relative dielectric constant

D. Mass of the two parallel plates

31. Biopotentials are ionic voltages produced when certain types of cells carry out electrochemical activities, true or false?

A. True

B. False

32. The biopotentials generated by the muscles of the heart with time is called?

A. Electroretinogram

B. Electroencephalogram

C. Electrocardiogram

D. Electrooculogram

32. All the following are examples of biopotentials, true or false?

- Electroencephalogram (EEG)
- Electromyogram (EMG)
- Electroretinogram (ERG)

A. True

B. False

33. How can you define electromyogram (EMG)?

A. The recorded representation of bioelectric potentials generated by the activity of the brain (Neuronal activity.

B. The biopotentials generated by the muscles of the heart with time.

C. The recorded representation of bioelectric potentials generated by the muscle activity.

D. The measure of the variations in the corneal-retinal potential.

34. Electroencephalogram (EEG) is used in the diagnosis of the:

A. Heart

B. Tumour

C. Neuromuscular diseases

D. Brain

35. Which of the following statements is true about Needle electrodes?

A. They are designed to measure bioelectric potentials near or within the cell.

B. They are designed to penetrate the skin so that they can record biopotentials like EEG signals from the brain.

C. They are designed to measure biopotentials from the surface of the skin.

D. Typical examples include the metal microelectrodes and micropipette.

36. The following are statements about heart sounds:

- **The beating of the heart and the pumping of the blood is associated with the generation of the sounds.**
- **The technique of listening to sounds by the heart and blood vessels is called auscultation.**
- **The physicians are trained to diagnose the heart disorders by listening to these sounds using a device called the stethoscope.**
- **A graphic record of the heart is called phonocardiogram.**

Are all the above statements are true or false?

A. True

B . False

37. The following are methods of blood flow measurement, except?

A. Magnetic blood flow measurement

B. Coriolis blood flow measurement

C. Ultrasonic blood flow measurement

D. Radiographic blood flow measurement

38. The following procedure refers a blood flow measurement method:

- **The dye is injected at a constant rate in the blood flow and a detector measures the dye concentration downstream.**
- **The concentration of the dye in the blood stream keep on increasing at lapses and it finally reaches a constant value.**
- **The blood flow = Injection of dye (milligrams/mm)/concentration of dye (milligrams per liter)**

Which blood flow measurement method is described by the above procedure?

A. Plethysmography chamber blood flow meter

B. Radiographic flow meter

C. Dye dilution flow meter

D. Ultrasonic blood flow meter

39. All the following are methods of blood pressure measurement, except?

A. Sphygmomanometer

B. Percutaneous method

C. Hagen-poiseuille analysis

D. Catheterization

41. Electrocardiograph (ECG) is a graphic recording or display of the time variant produced by the heart during the cardiac cycle, true or false?

A. True

B. False

42. What is Tidal volume with regard to lung capacities?

A. It is the extra volume that can be inspired.

B. It is the maximum volume of the gas that can be expelled from the lungs by forceful expiration after maximum inspiration.

C. It is the volume of gas inspired or expired during each respiration cycle.

D. It is the volume of air remaining in the lungs at the end of expiratory level.

43. What biological measurement is done by the Spirometer?

A. Blood pressure measurement

B. Blood Flow measurement

C. Respiratory volume measurements

D. Blood sugar measurement

44. What is the function of a Nebulizer as a respiratory therapy?

A. It is a device used to administer medication in the form of mist inhaled into the lungs and used for treatment of asthma, and other pulmonary disorders.

B. It is a device used to remove liquid or gases by suction from the body.

C. It is equipment used to increase humidity of the inspired air.

D. None of the above.

45. The following are properties of ultrasound waves as applied in medical imaging, except?

A. They are longitudinal

B. They are acoustic

C. They are electromagnetic

D. They depend upon the medium through which it propagates

46. Source of Bioelectric potential is _____ in nature.
a) electronic
b) electric
c) ionic
d) mechanical

Answer: c
Explanation: Bioelectric potentials are generated at cellular level and the source of these potentials is ionic in nature. The prominent ions are K^+, Na^+, and Cl^-. Electronic potential is seen in commonly used cells for example the Galvanic cell. Mechanical potential is found nowhere. Electrical potential is found in electricity.

47. Palsied muscles mean _____
a) paralyzed muscles
b) active muscles
c) voluntary muscles
d) involuntary muscles

Answer: a
Explanation: Palsied is an adjective that means paralyzed. It is used to describe a muscle on which an individual has lost all control (cannot move). Voluntary muscles are the muscles on which the individual has complete control. Involuntary muscles are the ones on which the individual has no control for example heart wall muscles.

48. The principal ion that is not involved with the phenomena of producing cell potentials is _____
a) sodium
b) potassium
c) chlorine
d) hydrogen

Answer: d
Explanation: Sodium (Na^+), Potassium (K^+), and Chlorine (Cl^-) are the principal ions involved with the phenomena of producing cell potentials. Na+ is present outside the cell membrane and creates a positively charged environment outside the cell

membrane. Cl^- is present inside the cell membrane and is responsible for the negative environment inside the cell membrane.

49. What is the relatively static membrane potential of quiescent cells called?
a) half-cell potential
b) action potential
c) resting membrane potential
d) cell potential

Answer: c
Explanation: Resting membrane potential or the resting potential is the relative static membrane potential of quiescent cell. That is if the resting membrane potential of a neuron is about -70 mV (mV=millivolt) it means that the inside of the neuron is 70 mV less than the outside of the neuron. An action potential occurs when the potential of the membrane of a given axonal position increases and decreases rapidly. This depolarization causes depolarization of adjacent positions in a similar way.

50. The variation of the electrical potential associated with the passage of a pulse along the membrane of a muscle cell or a nerve cell is called _____
a) muscle potential
b) action potential
c) resting potential
d) half-cell potential

Answer: b
Explanation: An action potential occurs when the potential of the membrane of a given axonal position increases and decreases rapidly. This depolarization causes depolarization of adjacent positions in a similar way. Resting membrane potential or the resting potential is the relative static membrane potential of a quiescent cell.

51. Cells depolarize and action potential in generated as soon as a stimulus is applied.
a) True
b) False

Answer: b
Explanation: This statement is False. This is because unless a stimulus above a certain minimum value is applied, the cell will not be depolarized and no action potential will

be generated. This value of potential above which the depolarizes and an action potential is generated is known as the stimulus threshold.

52. After a cell is stimulated, a finite period of time is required for the cell to return to its pre-stimulus state. This period is known as _____
a) restoration period
b) refactory period
c) regain period
d) regenerative period

Answer: b
Explanation: After a cell is stimulated, a finite period of time is required for the cell to return to its pre-stimulus state. This is because the energy associated with the action potential is developed from metabolic process within the cell which takes time for completion. This time period is called refactory period.

53. Electrooculography (EOG/E.O.G.) is a technique for measuring what?
a) abnormal function of the retina
b) heart rate
c) respiration rate
d) cornea-retinal standing potential

Answer: d
Explanation: Electrooculography (EOG / E.O.G) is a technique for measuring the potential of the corneal retinal standing potential that exists between the front and back of the human eye. The resulting signal is called electrooculogram. The main applications are in the diagnosis of ophthalmology and the recording of eye movements.

54. EKG stands for _____
a) Electrocardiography
b) Electroencephalography
c) Electromyography
d) Electrtokinetcography

Answer: a
Explanation: Electrocardiography (ECG or EKG) is the way toward recording the electrical action of the heart over some stretch of time utilizing anodes put on the skin. It could simply be understood as the electrical representation of heart beat. Electroencephalography is the electrical recording of brain.

55. Phonocardiography is listening to _____
a) arm muscle sound

b) lungs sound
c) heart sound
d) respiratory tract sound

Answer: c
Explanation: A phonocardiogram (or PCG) is a record high-constancy recording of sounds and mumble made by the heart with the assistance of the machine called phonocardiography. Consequently, phonocardiography is the chronicle of the considerable number of sounds made by the heart amid a heart cycle. Mostly stethoscope is used phonocardiography.

This set of Biomedical Instrumentation Multiple Choice Questions & Answers (MCQs) focuses on “Electrodes for EMG”.

56. Which of the following is a preferred electrode for measuring EMG?
a) surface electrodes
b) needle electrodes
c) pregelled electrodes
d) scalp electrodes

Answer: b
Explanation: Electrodes for EMG work are usually of the needle type. Needle electrodes are used in clinical EMG, neurography and other electrophysiological measurements of the muscle tissues underneath the skin. Surface electrodes are preferred to measure ECG. Scalp electrodes for EEG.

57. Generally what is the material of needle electrodes?
a) stainless steel
b) copper
c) lead
d) iron

Answer: a
Explanation: The material of the needle electrode is generally stainless steel. In spite of the face that stainless steel is unfavorable electrode material from the point of view of noise, it is preferred in EMG work. It is due to its mechanical solidity and low price.

58. Monopolar needle electrodes are having coatings of which material over the stainless steel wires which are bare only at the tips?
a) carbon
b) calcium
c) sodium

d) teflon

Answer: d
Explanation: The monopolar needle electrode consists of a teflon coated stainless steel wire. The wire is bare only at the tip. It is found that after the needle has been used a number of times, the teflon coating will recede, increasing the tip area. The needle should be discarded when this happens.

59. Which electrode can be used to pick up signals from individual fibers of muscle tissues?
a) biopolar needle electrode
b) concentric core needle electrode
c) multi-element needle electrode
d) monopolar needle electrode

Answer: c
Explanation: Multi-element needle electrodes are used to pick up the signals from individual fibers of muscle tissue. Special needles are available using 25-micron diameter electrode surfaces. They have up to 14 pickup surfaces down the side of one needle.

60. _____ instrument is used to hold patients head and guide the placement of electrodes.
a) Monotaxic
b) Stereotonic
c) Stereotaxic
d) Monotonic

Answer: c
Explanation: For measurement of potential from specific part of the brain, longer needles are inserted inside the head. The needles are precisely located by means of a map or atlas of the brain. A special instrument called stereotaxic instrument is used to hold the subject's head and guide the placement of the electrodes.

61. The ground electrode is usually positioned over which body structures?
a) bony
b) hairy
c) fleshy
d) sweaty

Answer: a
Explanation: The ground electrode is usually positioned over bony structures rather

than over large muscle masses, in the vicinity of the recording and stimulating electrodes, and where possible, equidistant from them. Hairy areas don't transmit proper signals. It increases impedance of the skin.

62. When intramuscular EMG is required to look into the electrical activities of deeper or overlaid muscles, _____ electrodes are used.
a) plate shape electrodes
b) surface electrodes
c) thin thread electrodes
d) fine wire electrodes

Answer: d
Explanation: When intramuscular EMG is required to look into the electrical activities of deeper or overlaid muscles, thin and flexible fine wire electrodes are used. These electrodes are inserted into the muscle site of interest. The needle or steel cannula is removed, and the electrode wires are connected to the steel spring adapters to minimize movement artefacts.

63. The contraction of the skeletal muscles results in the generation of action potential in the individual muscle fibers. Record of this action potential is called _____
a) ECG
b) EMG
c) EEG
d) EKG

Answer: b
Explanation: The contraction of the skeletal muscles results in the generation of action potential in the individual muscle fibers, a record of such action potential is known as electromyogram (EMG). The activity is similar to that observed in the cardiac muscles (ECG or EKG), but in the skeletal muscles, repolarization takes place much more rapidly. The action potential lasts for only few milliseconds.

64. In voluntary contraction of the skeletal muscles, the muscle potential ranges from _____
a) 50 uV – 5 mV
b) 50 mV – 5 V
c) 0.05 uV – 2 mV
d) 50 mV – 500 mV

Answer: a
Explanation: In voluntary contraction of the skeletal muscles, the muscle potential

ranges from 50 uV – 5 mV and duration from 2 to 15 ms. The values vary with the anatomic position of the muscle and the size and location of the electrode. In a relaxed muscle there are no action potentials.

65. Electrodes to measure EEG are placed on _____
a) forehead
b) scalp
c) cheek
d) ears

Answer: b
Explanation: Electrode to measure EEG are placed on the scalp. The position of each electrode is specified using the International 10/20 system. Each electrode site is labeled with a letter and a number.

66. According to the international 10/20 system to measure EEG, even number denotes which side of the brain?
a) left
b) top
c) bottom
d) right

Answer: d
Explanation: The position of each electrode is specified using the International 10/20 system. Each electrode site is labeled with a letter and a number. Even number denotes the right side of the head.

67. Letter F in the EEG electrode placement system denotes?
a) front
b) face
c) frontal lobe
d) fast

Answer: c
Explanation: F denotes frontal lobe of the head. The position of each electrode is specified using the International 10/20 system. Each electrode site is labeled with a letter and a number.

68. Normal EEG frequency range is _____
a) 50-500Hz
b) 0.5-50HZ
c) 0.05-5Hz

d) 1-200Hz

Answer: b
Explanation: The frequency varies greatly with different behavioral states. The normal EEG frequency content ranges from 0.5 to 50Hz. The nature of the wave varies over the different parts of the scalp.

69. The letter T in the EEG electrode placement system denotes?
a) temporal lope
b) temper lobe
c) trace
d) timpanic

Answer: a
Explanation: T denotes temporal lobe of the head. The position of each electrode is specified using the International 10/20 system. Each electrode site is labeled with a letter and a number.

70. According to the international 10/20 system to measure EEG, odd number denotes which side of the brain?
a) left
b) right
c) top
d) front

Answer: a
Explanation: The position of each electrode is specified using the International 10/20 system. Each electrode site is labeled with a letter and a number. Odd number denotes the left side of the head.

71. The delta wave in EEG ranges from _____
a) 0.5-4Hz
b) 4-8Hz
c) 8-13Hz
d) 13-22Hz

Answer: a
Explanation: The delta wave in EEG ranges from 0.5-4Hz. The theta wave in EEG ranges from 4-8Hz. The alpha wave in EEG ranges from 8-13Hz and beta from 13-22Hz.

72. Disturbance in the EEG pattern resulting from the external stimuli is called _____
a) provoked response

b) ckoored response
c) evoked response
d) impulse response

Answer: c
Explanation: Disturbance in the EEG pattern resulting from the external stimuli is called evoked response. The stimuli could be a flash light or a click of sound. The stimuli can be repeated and the EEG waveform can be observed to find the activities occurring because of the stimuli.

73. The peak to peak amplitude of the waves that can be picked from the scalp is _____

a) 100mV
b) 100V
c) 100uV
d) 10mV

Answer: c
Explanation: The EEG signal can be picked up with electrodes either from the scalp or directly from the cerebral cortex. The peak to peak amplitude of the waves that can be picked up from the scalp is normally 100uV. The frequency varies greatly with different behavioral states.

74. Which rhythm is the principal component of the EEG that indicates the alertness of the brain?

a) theta rhythm
b) gamma rhythm
c) beta rhythm
d) alpha rhythm

Answer: d
Explanation: The alpha rhythm is the principal component of the EEG and is an indicator of the state of alertness of the brain. It serves as an indicator of the depth of anesthesia in the operating room. The frequency of the EEG seems to be affected by the mental activity of a person.

TRANSISTOR

Q1. A transistor has ……

1. one pn junction
2. **two pn junctions**
3. three pn junctions
4. four pn junctions

Q2. The number of depletion layers in a transistor is ……

1. four
2. three
3. one
4. **two**

Q3. The base of a transistor is ……. doped

1. heavily
2. moderately
3. **lightly**
4. none of the above

Q4. The element that has the biggest size in a transistor is ……..

1. **collector**
2. base
3. emitter
4. collector-base-junction

Q5. In a pnp transistor, the current carriers are …….

1. acceptor ions
2. donor ions
3. free electrons
4. **holes**

Q6. The collector of a transistor is ……. doped

1. heavily
2. **moderately**
3. lightly
4. none of the above

Q7. A transistor is a …… operated device

1. **current**
2. voltage
3. both voltage and current
4. none of the above

Q8. In a npn transistor, ……. are the minority carriers

1. free electrons

2. **holes**
3. donor ions
4. acceptor ions

Q9. The emitter of a transistor is …… doped

1. lightly
2. **heavily**
3. moderately
4. none of the above

Q10. In a transistor, the base current is about …….. of emitter current

1. 25%
2. 20%
3. 35 %
4. **5%**

Q11. At the base-emitter junctions of a transistor, one finds ……

1. a reverse bias
2. a wide depletion layer
3. **low resistance**
4. none of the above

Q12. The input impedance of a transistor is …….

1. high
2. **low**
3. very high
4. almost zero

Q13. Most of the majority carriers from the emitter ……..

1. recombine in the base
2. recombine in the emitter
3. **pass through the base region to the collector**
4. none of the above

Q14. The current I_B is ……

1. **electron current**
2. hole current
3. donor ion current
4. acceptor ion current

Q15. In a transistor ……..

$I_C = I_E + I_B$

$I_B = I_C + I_E$

$I_E = I_C - I_B$

$I_E = I_C + I_B$

Q16. The value of α of a transistor is …….

- more than 1
- **less than 1**
- 1
- none of the above

Q17. $I_C = \alpha I_E +$ ……

1. I_B
2. I_{CEO}
3. **I_{CBO}**
4. βI_B

Q18. The output impedance of a transistor is ……..

1. **high**
2. zero
3. low
4. very low

Q19. In a tansistor, I_C = 100 mA and I_E = 100.2 mA. The value of β is ……

1. 100
2. 50
3. about 1
4. **200**

Q20. In a transistor if β = 100 and collector current is 10 mA, then I_E is ……

1. 100 mA
2. **100.1 mA**
3. 110 mA
4. none of the above

Q21. The relation between β and α is ……..

1. $\beta = 1 / (1 - \alpha)$
2. **$\beta = (1 - \alpha) / \alpha$**
3. **$\beta = \alpha / (1 - \alpha)$**
4. $\beta = \alpha / (1 + \alpha)$

Q22. The value of β for a transistor is generally ……..

1. 1
2. less than 1
3. **between 20 and 500**
4. above 500

Q23. The most commonly used transistor arrangement is …… arrangement

1. **common emitter**
2. common base
3. common collector
4. none of the above

Q24. The input impedance of a transistor connected in …….. arrangement is the highest

1. common emitter
2. **common collector**
3. common base
4. none of the above

Q25. The output impedance of a transistor connected in ……. arrangement is the highest

1. common emitter
2. common collector
3. **common base**
4. none of the above

Q26. The phase difference between the input and output voltages in a common base arrangement is …….

1. 180°
2. 90°
3. 270°
4. **0°**

Q27. The power gain in a transistor connected in ……. arrangement is the highest

1. **common emitter**
2. common base
3. common collector
4. none of the above

Q28. The phase difference between the input and output voltages of a transistor connected in common emitter arrangement is ……

1. **0°**
2. **180°**
3. 90°
4. 270°

Q29. The voltage gain in a transistor connected in ……. arrangement is the highest

1. common base
2. **common collector**
3. **common emitter**
4. none of the above

Q30. As the temperature of a transistor goes up, the base-emitter resistance ……

1. **decreases**
2. increases
3. remains the same
4. none of the above

Answer : 1

Q31. The voltage gain of a transistor connected in common collector arrangement is …….

1. equal to 1
2. more than 10
3. more than 100
4. **less than 1**

Q32. The phase difference between the input and output voltages of a transistor connected in common collector arrangement is ……

1. 180°
2. **0°**
3. 90°
4. 270°

Q33. $I_C = \beta I_B +$ …….

1. I_{CBO}
2. I_C
3. **I_{CEO}**
4. αI_E

Q34. $I_C = [\alpha / (1 - \alpha)] I_B +$ ……

1. **I_{CEO}**
2. I_{CBO}
3. I_C
4. $(1 - \alpha) I_B$

Q35. $I_C = [\alpha / (1 - \alpha)] I_B + [\ldots\ldots / (1 - \alpha)]$

1. **I_{CBO}**
2. I_{CEO}
3. I_C
4. I_E

Q36. BC 147 transistor indicates that it is made of …….

1. germanium
2. **silicon**
3. carbon
4. none of the above

Q37. I_{CEO} = (……) I_{CBO}

1. β
2. 1 + α
3. **1 + β**
4. none of the above

Q38. A transistor is connected in CB mode. If it is not connected in CE mode with same bias voltages, the values of I_E, I_B and I_C will …….

1. **remain the same**
2. increase
3. decrease
4. none of the above

Q39. If the value of α is 0.9, then value of β is ……..

1. 9
2. 0.9
3. 900
4. **90**

Q40. In a transistor, signal is transferred from a …… circuit

1. high resistance to low resistance
2. **low resistance to high resistance**
3. high resistance to high resistance
4. low resistance to low resistance

Q41. The arrow in the symbol of a transistor indicates the direction of …….

1. electron current in the emitter
2. electron current in the collector
3. **hole current in the emitter**
4. donor ion current

Q42. The leakage current in CE arrangement is ……. that in CB arrangement

1. **more than**
2. less than
3. the same as
4. none of the above

Q43. A heat sink is generally used with a transistor to ……

1. increase the forward current
2. decrease the forward current
3. compensate for excessive doping
4. **prevent excessive temperature rise**

Q44. The most commonly used semiconductor in the manufacture of a transistor is …….

1. germanium
2. **silicon**
3. carbon
4. none of the above

Q45. The collector-base junction in a transistor has ……..

1. forward bias at all times
2. **reverse bias at all times**
3. low resistance
4. none of the above

Q46. When transistors are used in digital circuits they usually operate in the …….

1. active region
2. breakdown region
3. **saturation and cutoff regions**
4. linear region

Q47. Three different Q points are shown on a dc load line. The upper Q point represents the …….

1. minimum current gain
2. intermediate current gain
3. maximum current gain
4. cutoff point

Q48. A transistor has a β_{DC} of 250 and a base current, I_B, of 20 μ A. The collector current, I_C, equals to …….

1. 500 μA
2. **5 mA**

3. 50 mA
4. 5 A

Q49. A current ratio of I_C/I_E is usually less than one and is called ……

1. beta
2. theta
3. **alpha**
4. omega

Q50. With the positive probe on an NPN base, an ohmmeter reading between the other transistor terminals should be ……

1. open
2. infinite
3. **low resistance**
4. high resistance

Q51. In a CE configuration, an emitter resistor is used for ……

1. **stabilization**
2. ac signal bypass
3. collector bias
4. higher gain

Q52. Voltage-divider bias provides …….

1. an unstable Q point
2. **a stable Q point**
3. a Q point that easily varies with changes in the transistor's current gain
4. a Q point that is stable and easily varies with changes in the transistor's current gain

Q53. To operate properly, a transistor's base-emitter junction must be forward biased with reverse bias applied to which junction?

1. collector-emitter
2. base-collector
3. base-emitter
4. **collector-base**

Q54. The ends of a load line drawn on a family of curves determine ……

1. **saturation and cutoff**
2. the operating point
3. the power curve
4. the amplification factor

Q55. If V_{CC} = +18 V, voltage-divider resistor R_1 is 4.7 kΩ, and R_2 is 1500 Ω, then the base bias voltage is …….

1. 8.7 V
2. **4.35 V**
3. 2.9 V
4. 0.7 V

Q56. The C-B configuration is used to provide which type of gain?

1. **voltage**
2. current
3. resistance
4. power

Q57. The Q point on a load line may be used to determine ……

1. V_C
2. V_{CC}
3. **V_B**
4. I_C

Q58. A transistor may be used as a switching device or as a ……

1. fixed resistor
2. tuning device
3. rectifier
4. **variable resistor**

Q59. If an input signal ranges from 20–40 μA (microamps), with an output signal ranging from .5–1.5 mA (milliamps), what is the ac beta?

1. 0.05
2. 20
3. **50**
4. 500

Q60. Beta's current ratio is ……..

1. **I_C/I_B**
2. I_C/I_E
3. I_B/I_E
4. I_E/I_B

Q61. A collector characteristic curve is a graph showing ……..

1. emitter current (I_E) versus collector-emitter voltage (V_{CE}) with (V_{BB}) base bias voltage held constant
2. **collector current (I_C) versus collector-emitter voltage (V_{CE}) with (V_{BB}) base bias voltage held constant**

3. collector current (I_C) versus collector-emitter voltage (V_C) with (V_{BB}) base bias voltage held constant
4. collector current (I_C) versus collector-emitter voltage (V_{CC}) with (V_{BB}) base bias voltage held constant

Q62. With low-power transistor packages, the base terminal is usually the …….

1. tab end
2. **middle**
3. right end
4. stud mount

Q63. When a silicon diode is forward biased, V_{BE} for a CE configuration is ……..

1. voltage-divider bias
2. 0.4 V
3. **0.7 V**
4. emitter voltage

Q64. What is the current gain for a common-base configuration where I_E = 4.2 mA and I_C = 4.0 mA?

1. 16.8
2. 1.05
3. 0.2
4. **0.95**

Q65. With a PNP circuit, the most positive voltage is probably ……

1. **ground**
2. V_C
3. V_{BE}
4. V_{CC}

Q66. If a 2 mV signal produces a 2 V output, what is the voltage gain?

1. 0.001
2. 0.004
3. **100**
4. **1000**

Q67. Most of the electrons in the base of an NPN transistor flow ……

1. out of the base lead
2. **into the collector**
3. into the emitter
4. into the base supply

Q68. In a transistor, collector current is controlled by ……..

1. collector voltage
2. **base current**
3. collector resistance
4. all of the above

Q69. Total emitter current is ……

1. $I_E - I_C$
2. $I_C + I_E$
3. **$I_B + I_C$**
4. $I_B - I_C$

Q70. Often a common-collector will be the last stage before the load; the main function(s) of this stage is to …….

1. provide voltage gain
2. provide phase inversion
3. provide a high-frequency path to improve the frequency response
4. **buffer the voltage amplifiers from the low-resistance load and provide impedance matching for maximum power transfer**

Q71. For a CC configuration to operate properly, the collector-base junction should be reverse biased, while forward bias should be applied to …… junction.

1. **collector-emitter**
2. base-emitter
3. collector-base
4. cathode-anode

Q72. The input/output relationship of the common-collector and common-base amplifiers is ……..

1. 270 degrees
2. 180 degrees
3. 90 degrees
4. **0 degrees**

Q73. If a transistor operates at the middle of the dc load line, a decrease in the current gain will move the Q point …….

1. off the load line

2. nowhere
3. up
4. **down**

Q74. Which is the higher gain provided by a CE configuration?

1. voltage
2. current
3. resistance
4. **power**

Q75. What is the collector current for a CE configuration with a beta of 100 and a base current of 30 μA?

1. 30 μA
2. 0.3 μA
3. **3 mA**
4. 3 MA

……..

ICs

Q1. An IC has ……. size

1. Very large
2. Large
3. **Extremely small**
4. None of the above

Q2. ICs are generally made of ……

1. **Silicon**
2. Germanium
3. Copper
4. None of the above

Q3. …… ICs are the most commonly used

1. Thin films
2. **Monolithic**
3. Hybrid
4. None of the above

Q4. The most popular form of IC package is …….

1. DIL
2. **Flatpack**
3. TO-5
4. None of the above

Q5. ……. cannot be fabricated on an IC

1. Transistors
2. Diodes
3. Resistors
4. **Large inductors and transformers**

Q6. An audio amplifier is an example of ……

1. Digital IC
2. **Linear IC**
3. Both digital and linear IC
4. None of the above

Q7. The active components in an IC are …….

1. Resistors
2. Capacitors
3. **Transistors and diodes**
4. None of the above

Q8. We use ……. ICs in computers

1. **Digital**
2. Linear
3. Both digital and linear
4. None of the above

Q9. The SiO2 layer in an IC acts as …….

1. A resistor
2. **An insulating layer**
3. Mechanical output
4. None of the above

Q10. ICs are used in ……

1. Linear devices only
2. Digital devices only
3. **Both linear and digital devices**
4. None of the above

Q11. A transistor takes …… inductor on a silicon IC chip

1. **Less space than**
2. More space than

3. Same space as
4. None of the above

Q12. The most popular types of ICs are ……..

1. Thin-film
2. Hybrid
3. **Thick-film**
4. **Monolithic**

Q13. Digital ICs process …….

1. Linear signals only
2. **Digital signals only**
3. **Both digital and linear signals**
4. None of the above

Q14. Operational amplifiers use ……..

1. Linear ICs
2. **Digital ICs**
3. Both linear and digital ICs
4. None of the above

Q15. Which of the following is most difficult to fabricate in an IC?

1. Diode
2. Transistor
3. FET
4. **Capacitor**

……

DIODES

Q1. A crystal diode has ……

1. **one pn junction**
2. two pn junctions
3. three pn junctions
4. none of the above

Q2. A crystal diode has forward resistance of the order of ……

1. kΩ
2. **Ω**
3. MΩ
4. none of the above

Q3. If the arrow of crystal diode symbol is positive w.r.t. bar, then diode is ……. biased.

1. **forward**
2. reverse
3. either forward or reverse
4. none of the above

Q4. The reverse current in a diode is of the order of …….

1. kA
2. mA
3. **µA**
4. A

Q5. The forward voltage drop across a silicon diode is about ……

1. 2.5 V
2. 3 V
3. 10 V
4. **0.7 V**

Q6. A crystal diode is used as ……

1. an amplifier
2. **a rectifier**
3. an oscillator
4. a voltage regulator

Q7. The d.c. resistance of a crystal diode is ……. its a.c. resistance

1. the same as
2. more than
3. **less than**
4. none of the above

Q8. An ideal crystal diode is one which behaves as a perfect …….. when forward biased.

1. **conductor**
2. insulator
3. resistance material
4. none of the above

Q9. The ratio of reverse resistance and forward resistance of a germanium crystal diode is about …….

1. 1 : 1
2. 100 : 1
3. 1000 : 1

4. **40,000 : 1**

Q 10. The leakage current in a crystal diode is due to …….

1. **minority carriers**
2. majority carriers
3. junction capacitance
4. none of the above

Q11. If the temperature of a crystal diode increases, then leakage current ……..

1. remains the same
2. decreases
3. **increases**
4. becomes zero

Q12. The PIV rating of a crystal diode is …….. that of equivalent vacuum diode

1. the same as
2. **lower than**
3. more than
4. none of the above

Q13. If the doping level of a crystal diode is increased, the breakdown voltage…….

1. remains the same
2. is increased
3. **is decreased**
4. none of the above

Q14. The knee voltage of a crystal diode is approximately equal to …….

1. applied voltage
2. breakdown voltage
3. forward voltage
4. **barrier potential**

Q15. When the graph between current through and voltage across a device is a straight line, the device is referred to as …….

1. **linear**
2. active
3. nonlinear
4. passive

Q16. When the crystal current diode current is large, the bias is ……

1. **forward**
2. inverse

3. poor
4. reverse

Q17. A crystal diode is a …… device

1. **non-linear**
2. bilateral
3. linear
4. none of the above

Q18. A crystal diode utilises …….. characteristic for rectification

1. reverse
2. **forward**
3. forward or reverse
4. none of the above

Q19. When a crystal diode is used as a rectifier, the most important consideration is ……..

1. forward characteristic
2. doping level
3. reverse characteristic
4. **PIC rating**

Q20. If the doping level in a crystal diode is increased, the width of depletion layer……..

1. remains the same
2. is decreased
3. **in increased**
4. none of the above

Q21. A zener diode has ……..

1. **one pn junction**
2. two pn junctions
3. three pn junctions
4. none of the above

Q22. A zener diode is used as …….

1. an amplifier
2. **a voltage regulator**
3. a rectifier
4. a multivibrator

Q23. The doping level in a zener diode is …… that of a crystal diode

1. the same as
2. less than

3. **more than**
4. none of the above

Q24. A zener diode is always …… connected.

1. **reverse**
2. forward
3. either reverse or forward
4. none of the above

Q25. A zener diode utilizes …….. characteristics for its operation.

1. forward
2. **reverse**
3. both forward and reverse
4. none of the above

Q26. In the breakdown region, a zener didoe behaves like a …… source.

1. **constant voltage**
2. constant current
3. constant resistance
4. none of the above

27. A zener diode is destroyed if it……..

1. is forward biased
2. is reverse biased
3. **carrier more than rated current**
4. none of the above

Q28. A series resistance is connected in the zener circuit to……..

1. properly reverse bias the zener
2. **protect the zener**
3. properly forward bias the zener
4. none of the above

Answer :2

A29. A zener diode is ……. device

1. **a non-linear**
2. a linear
3. an amplifying
4. none of the above

Q30. A zener diode has …….. breakdown voltage

1. undefined
2. **sharp**
3. zero

4. none of the above

Q31. ……. rectifier has the lowest forward resistance

1. **solid state**
2. vacuum tube
3. gas tube
4. none of the above

Q32. Mains a.c. power is converrted into d.c. power for …….

1. lighting purposes
2. heaters
3. **using in electronic equipment**
4. none of the above

Q33. The disadvantage of a half-wave rectifier is that the…….

1. components are expensive
2. diodes must have a higher power rating
3. **output is difficult to filter**
4. none of the above

Q34. If the a.c. input to a half-wave rectifier is an r.m.s value of 400/√2 volts, then diode PIV rating is …….

1. 400/√2 V
2. **400 V**
3. 400 x √2 V
4. none of the above

Answer :2

Q35. The ripple factor of a half-wave rectifier is ……

1. 2
2. **1.21**
3. 2.5
4. 0.48

Q36. There is a need of transformer for …….

1. half-wave rectifier
2. **centre-tap full-wave rectifier**
3. bridge full-wave rectifier
4. none of the above

Q37. The PIV rating of each diode in a bridge rectifier is …… that of the equivalent centre-tap rectifier

1. **one-half**
2. the same as

3. twice
4. four times

Q38. For the same secondary voltage, the output voltage from a centre-tap rectifier is …….. than that of bridge rectifier

1. twice
2. thrice
3. **four time**
4. **one-half**

Q39. If the PIV rating of a diode is exceeded, ……

1. the diode conducts poorly
2. **the diode is destroyed**
3. the diode behaves like a zener diode
4. none of the above

Q40. A 10 V power supply would use ……. as filter capacitor.

1. paper capacitor
2. mica capacitor
3. **electrolytic capacitor**
4. air capacitor

Q41. A 1,000 V power supply would use …….. as a filter capacitor

1. **paper capacitor**
2. air capacitor
3. mica capacitor
4. electrolytic capacitor

Q42. The …….. filter circuit results in the best voltage regulation

1. **choke input**
2. capacitor input
3. resistance input
4. none of the above

Q43. A half-wave rectifier has an input voltage of 240 V r.m.s. If the step-down transformer has a turns ratio of 8:1, what is the peak load voltage? Ignore diode drop.

1. 27.5 V
2. 86.5 V
3. 30 V
4. **42.5 V**

Q44. The maximum efficiency of a half-wave rectifier is …….

1. **40.6 %**

2. 81.2 %
3. 50 %
4. 25 %

Q45. The most widely used rectifier is …….

1. half-wave rectifier
2. centre-tap full-wave rectifier
3. **bridge full-wave rectifier**
4. none of the above

……

CAPACITORS

Q 1. A sine wave voltage is applied across a capacitor. When the frequency of the voltage is decreased, the current

(A) Increases

(B) Remains constant

(C) Decreases

(D) Ceases

Q 2. Two series capacitors (one 2 μF, the other of unknown value) are charged from a 24 V source. The 2 μF capacitor is charged to 16 V and the other to 8 V. The value of the unknown capacitor is

(A) 1 μF

(B) 2 μF

(C) 4 μF

(D) 8 Mf

Q 3. Total parallel capacitance is less than that of the smallest capacitor in parallel.

(A) True

(B) False

Q 4. A 2 μF, a 4 μF, and a 10 μF capacitor are connected in series. The total capacitance is less than

(A) 2 μF
(B) 4 μF
(C) 10 μF
(D) 1.5 μF

Q 5. XC is inversely proportional to frequency and capacitance.
(A) True
(B) False

Q 6. An uncharged capacitor and a 1 kΩ resistor are connected in series with a switch and a 6 V battery. At the instant the switch is closed, the voltage across the capacitor is
(A) 6 V
(B) 3 V
(C) 2 V
(D) 0 V

Q 7. A capacitor blocks A.C. but passes D.C.
(A) True
(B) False

Q 8. A 220 Ω resistor is in series with a 2.2 μF capacitor. The time constant is
(A) 48 μs
(B) 480 μs
(C) 2.42 μs
(D) 24 μs

Q 9. The dielectric constant is the ability of a material to establish an electric field.
(A) True
(B) False

Q 10. A capacitor of 0.02 μF is larger than
(A) 0.000020 F
(B) 200,000 pF

(C) 2,000 pF
(D) All of the above

Q 11. A 4.7 MΩ resistor is in series with a 0.015 µF capacitor. The combination is across a 12 V source. How long does it take the capacitor to fully charge?
(A) 35 ms
(B) 352 ms
(C) 3.5 s
(D) 70.5 ms

Q 12. Two 0.68 µF capacitors are connected in series across a 10 kHz sine wave signal source. The total capacitive reactance is
(A) 46.8 Ω
(B) 4.68 Ω
(C) 23.41 Ω
(D) 11.70 Ω

Q 13. Capacitance is directly proportional to plate area and plate separation.
(A) True
(B) False

Q 14. A 10 µF, 20 µF, 22 µF, and 100 µF capacitor are in parallel. The total capacitance is
(A) 2.43 µF
(B) 4.86 µF
(C) 100 µF
(D) 152 µF

Q 15. Voltage leads current by 90° in a capacitor.
(A) True
(B) False

Q 16. A capacitor of 2,000 pF is greater than
(A) 0.02 µF
(B) 0.002 µF

(C) 0.0002 μF
(D) 0.0000002 F

Q 17. The true power in a capacitor is zero.
(A) True
(B) False

Q 18. A 12 kHz voltage is applied to a 0.33 μF capacitor, and 200 mA of rms current is measured. What is the value of the voltage?

(A) 8 V
(B) 80 V
(C) 80 mV
(D) 800 mV

Q 19. The ohm is the unit of capacitance.
(A) True
(B) False

Q 20. When the plate area of a capacitor increases,
(A) The capacitance increases
(B) The capacitance decreases
(C) The capacitance is unaffected
(D) The voltage it can withstand increases

Q 21. When the voltage across a capacitor is tripled, the stored charge
(A) Triples
(B) Is cut to one-third
(C) Stays the same
(D) Doubles

Q 22. A 0.47 μF capacitor is across a 2 kHz sine wave signal source. The capacitive reactance is
(A) 170 Ω
(B) 17 Ω
(C) 0.000169 Ω
(D) 1.7 Ω

Q 23. Four 0.15 µF capacitors are in parallel. The total capacitance is
(A) 0.15 µF
(B) 0.30 µF
(C) 0.6 µF
(D) 0.8 µF

Q 24. Which of the following capacitors is polarized?
(A) Mica
(B) Ceramic
(C) Plastic-film
(D) Electrolytic

Answer: Option D
Q 25. Energy is stored by a capacitor in a magnetic field.
(A) True
(B) False

Q 26. Of the following capacitors, which one has the highest dielectric constant?
(A) Air
(B) Mica
(C) Glass
(D) Paper

Q 27. A 0.01 µF capacitor equals how many picofarads?
(A) 100 pF
(B) 1,000 pF
(C) 10,000 pF
(D) 100,000 pF

Q 28. A capacitor and a resistor are connected in series to a sine wave generator. The frequency is set so that the capacitive reactance is equal to the resistance and, thus, an equal amount of voltage appears across each component. If the frequency is increased
(A) VR > VC
(B) VC > VR

(C) VR = VC
(D) VR and VC = 0

Q 29. A 0.00022 F capacitor equals how many microfarads?
(A) 22 μF
(B) 220 μF
(C) 2,200 μF
(D) 22,200 μF

Q 30. A 330 pF capacitor and a 220 pF capacitor are each connected across a 6 V dc source. The voltage across the 330 pF capacitor is
(A) 3 V
(B) 6 V
(C) 4 V
(D) 0 V

Q 31. Five time constants are required for a capacitor to charge fully or discharge fully.
(A) True
(B) False

Q 32. What is the capacitance when Q = 60 μC and V = 12 V?
(A) 720 μF
(B) 5 μF
(C) 50 μF
(D) 12 μF

MEASUREMENT AND INSTRUMENTATION

Q1. ……. are integrating instruments?

1. Ammeters
2. Voltmeters
3. Wattmeters

4. **Ampere-hour and Watt-hour meters**

Q2. Resistances can be measured with the help of a ……

1. Wattmeter
2. voltmeter
3. ammeter
4. **ohmmeter and resistance bridge**
5. all of the above

Q3. ……. instruments indicate the instantaneous value of the electrical quantity being measured at the time at which it is being measured?

1. Absolute
2. **Indicating**
3. Recording
4. Integrating

Q4. The use of ……. instruments is merely confined within laboratories as standardizing instruments.

1. **absolute**
2. indicating
3. recording
4. integrating
5. none of the above

Q5. ……. instruments measure the total quantity of electricity delivered at a particular time.

1. absolute
2. indicating
3. recording
4. **integrating**

Q6. According to application, instruments can be classified into ……. and …….

1. switch board
2. portable
3. **both 1 and 2**
4. moving coil
5. moving iron
6. both 4 and 5

Q7. The spring material used in a spring control device should have the following property:

1. should be non-magnetic
2. should have low-temperature co-efficient

3. should have low specific resistance
4. should not be subjected to fatigue
5. **all of the above**

Q8. Which of the following properties a damping oil must possess?

1. must be a good insulator
2. should be non-evaporating
3. should not have corrosive action upon the metal of the vane
4. the viscosity of the oil should not change with the temperature
5. **all of the above**

Q9. A ……. device prevents the oscillation of the moving system and enables the latter to reach its final position quickly.

1. deflecting
2. controlling
3. **damping**
4. all of the above

Q10. An induction meter can handle current upto …….

1. 10 A
2. 30 A
3. 60 A
4. **100 A**

Q11. For handling greater currents induction wattmeter are used in conjunction with …….

1. potential transformer
2. **current transformer**
3. power transformer
4. all of the above

Q12. …….. devices may be used for extending the range of the instruments?

1. shunts
2. multipliers
3. current transformers
4. potential transformer
5. **all of the above**

Q13. A moving-coil permanent-magnet instrument can be used as flux-meter by ……

1. using a low resistance shunt
2. using a high series resistance

3. **eliminating the control springs**
4. making control springs of large moment of inertia

Q14. A moving-coil permanent-magnet instrument can be used as …….. by using a low resistance shunt.

1. **ammeter**
2. voltmeter
3. flux-meter
4. ballistic galvanometer

Q15. A potentiometer may be used for ……

1. measurement of resistance
2. measurement of current
3. calibration of ammeter
4. calibration of voltmeter
5. **all of the above**

Q16. ……. are not used on D.C. circuits.

1. Mercury motor meters
2. commutator motor meters
3. **induction meters**
4. none of the above

Q17. ……. is an essential part of a motor meter .

1. an operating torque system
2. a barking device
3. revolution registering device
4. **all of the above**

Q18. Induction type single phase energy meters measure electric energy in ……..

1. kW
2. Wh
3. **kWh**
4. VAR
5. none of the above

Q19. The pointer of an indicating instrument should be ……

1. **very light**
2. very heavy
3. either 1 or 2
4. neither 1 nor 2

Q20. The household energy meter is ……..

1. an indicating instrument

2. a recording instrument
3. **an integrating instrument**
4. none of the above

Q21. The chemical effect of current is used in …….

1. **D.C. ammeter hour meter**
2. D.D. ammeter
3. D.C. energy meter
4. none f the above

Q22. In majority of instruments, damping is provided by ……

1. fluid friction
2. spring
3. **eddy currents**
4. all of the above

Q23. An ammeter is a …….. instrument.

1. **secondary instrument**
2. absolute instrument
3. recording instrument
4. integrating instrument

Q24. The switchboard instruments should be mounted in …….. position.

1. **vertical**
2. horizontal
3. either 1 or 2
4. neither 1 nor 2

Q25. The function of shunt in an ammeter is to ……..

1. **by pass the current**
2. increase the sensitivity of the ammeter
3. increase the resistance of ammeter
4. none of the above

Q26. The multiplier and the meter-coil in a voltmeter are in …….

1. **series**
2. parallel
3. series-parallel
4. none of the above

Q27. A moving iron instrument can be used for ……..

1. D.C. only
2. A.C. only
3. **both D.C. and A.C.**

4. none of the above

Q28. The scale of a rectifier instrument is ……..

1. **linear**
2. non-linear
3. either 1 or 2
4. neither 1 nor 2

Q29. To measure current at high frequency, we must use ……..

1. moving iron instrument
2. electrostatic instrument
3. **thermocouple instrument**
4. none of the above

Q30. The resistance in the circuit of the moving coil of a dynamometer should be ……

1. zero
2. low
3. **high**
4. none of the above

Q31. A dynamometer wattmeter can be used for ……

1. D.C. only
2. A.C. only
3. **both D.C. and A.C.**
4. none of the above

Q32. An induction wattmeter can be used for ……

1. **D.C. only**
2. A.C. only
3. both D.C. and A.C.
4. none of the above

Q33. In a 3-phase power measurement by two wattmeter method, the reading on one of the wattmeter is zero. The power factor of the load must be …….

1. unity
2. **0.5**
3. 0.3
4. zero

Q34. The adjustment of position of shading bands, in an energy meter is done to provide …….

1. **friction compensation**
2. creep compensation

3. braking torque
4. none of the above

Q35. An ohmmeter is a …… instrument.

1. moving iron
2. **moving coil**
3. dynamometer
4. none of the above

Q36. To measure a very high resistance, we should use ……..

1. Kelvin's double bridge
2. Wheat stone bridge
3. **Meggar**
4. none of the above

Q37. The electric power t a Meggar is provided by ……..

1. battery
2. **permanent magnet D.C. generator**
3. A.C. generator
4. any of the above

Q38. In a Meggar cntrolling torque is providec by …….

1. spring
2. gravity
3. **coil**
4. eddy current

Q39. The operating voltage of a Meggar is about …….V.

1. 6 V
2. 12 V
3. **40 V**
4. 100 V

Q40. Murray loop test can be used for location of …….

1. ground fault on a cable
2. short circuit fault on a cable
3. **both the ground fault and short-circuit fault**
4. none of the above

Q41. …… should be used for an accurate measurement of low D.C. voltage.

1. small range moving coil voltmeter
2. **D.C. potentiometer**
3. small range thermocouple voltmeter
4. none of the above

Q42. The best device to measure the true open circuit e.m.f. of a battery is ……

1. D.C. voltmetr
2. ammeter and a known resistance
3. **D.C. potentiometer**
4. none of the above

Q43. A phase shifting transformer is used in conjunction with ……..

1. D.C. potentiometer
2. **Drysdale potentiometer**
3. A.C. co-ordinate potentiometer
4. Crompton potentiometer

Q44. In order to achieve high accuracy, the slide wire of a potentiometer should be ……

1. **as long as possible**
2. as short as possible
3. neither too small nor too large
4. very thick

Q45. The stator of phase shifting transformer for use in conjunction with an A.C. potentiometer usually has a …….

1. single-phase winding
2. **two-phase winding**
3. three-phase winding
4. none of the above

Q46. In an A.C. co-ordinate potentiometer, the currents in the phase and quadrature potentiometer are adjusted to be ……..

1. **out of phase by 90°**
2. out of phase by 180°
3. out of phase by 60°
4. out of phase by 0°

Q48. For measurements on high voltage capacitors, the suitable bridge is …….

1. Wein bridge
2. Modified De Santy's bridge
3. **Schering bridge**
4. none of the above

Q50. Mutual inductance can be measured by using ……

1. Anderson bridge

2. Maxwell's bridge
3. **Heaviside bridge**
4. any of the above

Q51. The full range of audibility in audio frequency oscillator is ……

1. 0 to 20 Hz
2. **20 Hz to 2 kHz**
3. 20 Hz to 20 kHz
4. 20 Hz to 20 MHz

Q52. A liquid crystal display requires …….

1. **An AC drive**
2. A DC drive
3. Both AC and DC drive
4. None of these

Q53. The detectors used in optical sensors are …….

1. Photodiodes
2. Phototransistors
3. Laser
4. **Only (1) and (2)**
5. All the above

Q54. Optical sensors used for the displacement measurement works on the principal that …….

1. Intensity of light increases with distance
2. **Intensity of light decreases with distance**
3. Intensity of light remains constant with distance
4. Intensity of light increases with time

Q55. Capacitance sensor can measure very small displacement. It can be formed by varying ……

1. Separation
2. Area
3. Permittivity
4. **Either (1) or (2) or (3)**

Q56. The ionization gauge an instrument used for the measurement of …….

1. **Very low pressure**
2. Medium pressure
3. High pressure
4. Very high pressure

Q57. Dead weight gauge is used for the measurement of pressure of ……

1. About 1000 bar
2. About 2000 bar
3. About 5000 bar
4. **About 7000 bar**

Q59. The resistances of potential transformer winding is minimized by using …….

1. **Thick conductors and small length of turns**
2. Thin conductors and small length of turns
3. Thin conductors and large length of turns
4. Thick conductors and large length of turns

Q60. The nominal ratio for a current transformer is given by ……

1. **rated primary winding current/rated secondary winding current**
2. number of turns in the primary winding/number of turns in the secondary winding
3. number of turns in the secondary winding/number of turns in the primary winding
4. rated secondary winding current/rated primary winding current

Q61. Moving iron power factor meter are suitable for 3 phase balanced circuits. It consists of ……..

1. One control spring
2. Two control spring
3. Three control spring
4. **No control spring**

Ans. 4

Q62. If the frequency of electrodynamic power factor meter is doubled then its reading will become……

1. Twice of the original reading
2. Half of the original reading
3. Four times of the original reading
4. **Remains unaffected**

Q63. A moving iron frequency meter consists of …….

1. Two inductive circuits connected in parallel
2. **One inductive and one non inductive circuit connected in parallel**
3. Two non inductive circuits connected in parallel
4. One inductive and one non inductive circuit connected in series

Q64. The electrodynamic frequency meters have …….

1. **Linear scale and their readings does not depends on voltage**
2. Linear scale and their readings depends on voltage
3. Non linear scale and their readings does not depends on voltage
4. Non linear scale and their readings depends on voltage

Q65. The earth resistance can be measured by ……..

1. Fall of potential method
2. Using an earth tester
3. Ducter ohmmeter method
4. **Only (1) and (2)**
5. All the above

Q66. Electrical equipments are generally earthed through an electrode to avoid shocks when someone touches the body of the equipment. The earth resistance is effected by ……..

1. Depth of electrodes buried in the soil
2. Shape and material of earth electrodes
3. Specific resistance of the soil surrounding the electrode
4. **All of these**
5. None of these

Ans. 4

Q67. When a voltmeter – ammeter method is applied for the measurement of resistance, the voltmeter reads a value of 8.28 V and the ammeter reading is 4.14 mA. Then the value of the resistance will be ……

1. 2 kΩ
2. 20 kΩ
3. **200 kΩ**
4. 2000 kΩ

Q68. In AC bridges, the Wagner earth devices are used to …….

1. Remove all the earth capacitances from the bridge circuit
2. Remove harmonics
3. Reduce error caused by stray electric field
4. **All of these**

Q69. Shielding of the capacitor is done to ……

1. Make the value of capacitor definite
2. Balance the bridge without any problem
3. **Both (1) and (2)**
4. None of these

Q70. The indicating instruments with linear scale is …….

1. PMMC
2. Electrostatic instrument
3. Dynamometer instrument
4. Thermocouple instrument

Q71. The resistance potential divider method and capacitance potential divider method is used for……

1. Both AC and DC
2. **Former method can be used for both AC and DC and the later method can be used only for AC**
3. Former method can be used for AC only and the later method can be used for both AC and DC
4. Former method can be used for DC only and the later method can be used only for AC

Q72. The range of electrostatic voltmeter can be extended by using ……..

1. Resistance potential divider method
2. Capacitance potential divider method
3. **Both (1) and (2)**
4. None of these

Q73. The multiplying factor of electrostatic voltmeters is given by ……

1. **$(C + C_v) / C$**
2. $(C + C_v) / C_v$
3. $C / (C + C_v)$
4. $C_v / (C + C_v)$

Q74. In electrostatic instruments iron is not used for construction. These instruments are ……..

1. Free from hysteresis and eddy current losses
2. Free from temperature errors
3. Dependent on temperature errors
4. **Both (1) and (2)**
5. None of the above

Q75. If an electrostatic voltmeter is used on AC circuit and has non uniform waves, then it will read ……..

1. Average values
2. **RMS values**
3. Peak values
4. All of these

Q76. Electrostatic voltmeter instruments are suitable for ……

1. AC work only
2. DC work only
3. **Both AC and DC work**
4. None of these

Q77. A Kelvin's multicellular voltmeter has a torsion head and a coach spring for …….

1. Protection against accidental fraction of suspension due to vibration
2. For zero adjustment
3. **Torsion head for zero adjustment and coach spring for Protection against accidental fraction of suspension due to vibration**
4. Torsion head for Protection against accidental fraction of suspension due to vibration and coach spring for zero adjustment

Q78. In electrostatic voltmeters, the principle of their operation is the force of attraction between electric charges on neighboring plates between which potential difference is maintained. The attracted – disc type electrostatic instruments are used for the measurement of ……..

1. Very low voltages
2. Low voltages
3. High voltages
4. **Very high voltages**

Q79. Electrostatic instruments are generally used as ……

1. **Voltmeters**
2. Ammeters
3. Wattmeters
4. Watt-hour meters

Q80. If the quantity to be measured remains constant during the process of taking the repeated measurements then the random errors can be eliminated by ……

1. Calculating the mean of the number of repeated measurements
2. Calculating the median of the number of repeated measurements
3. Calculating the sum of the numbers of repeated measurements
4. **Either (1) or (2)**

Q81. The error between the mean of finite data set and mean of infinite data set is known as …….

1. True error of the mean
2. **Standard error of the mean**
3. Finite error

4. Infinite error

Q82. In a measurement system,

1. A single measurement components may have both random errors and systematic errors
2. A measurement system consists of several components with each component having separate errors
3. **Both the statement (1) & (2) are true**
4. Neither statement (1) nor statement (2) is true

Q83. If the degree of damping of an instrument should be adjusted to a value which is sufficient to enable the pointer to rise quickly to its deflected position without overshooting is called as ……

1. Overdamped
2. **Dead beat**
3. Underdamped
4. None of these

Q84. Due to overdamping, the instrument will become …….

1. Slow
2. Lethargic
3. Fast
4. **Both (1) and (2)**

Q85. While measuring resistance by the voltmeter – ammeter method, the maximum possible percentage error in the voltmeter and ammeter are ± 1.8% and ± 1.2% respectively. Then the maximum possible percentage error in the value of resistance will be ……..

1. ± 3%
2. ± 4%
3. ± 4.2%
4. ± 4.8%

Q86. If the resistance in a circuit is given by 80 Ω ± 0.2% and the current flowing through it is 5A ± 0.1%, then the uncertainty in the power will be …….

1. ± 0.2 %
2. **± 0.4 %**
3. ± 0.6 %
4. ± 0.8 %

Q87. When a 100 V moving iron voltmeter is of accuracy class 1 – 0 is used in a circuit, it reads 50 V. Then the maximum possible percentage error in the reading is ……

1. 1 %
2. **2 %**
3. 2.5 %
4. 3 %

Q88. In liquid crystal displays, the liquid crystal exhibits properties of ……

1. Liquid
2. Solids
3. Gases
4. **Both (1) and (2)**

Q89. In light emitting diode, the available light emitting region is ……..

1. Less than 2.5 mm
2. **From 2.5 to 25 mm**
3. Greater than 25 mm
4. Greater than 50 mm

Q90. Resolver works on the principal of mutual inductance variation. It is mainly used for the measurement of ……

1. Linear displacement
2. Non – linear displacement
3. **Rotary motion**
4. All of these

Q91. In rotary variable differential transformer, the mutual inductance between the primary and secondary coils varies ……

1. **Linearly with the angular displacement**
2. Non – linearly with the angular displacement
3. Linearly with the linear displacement
4. Non – linearly with the linear displacement

Q92. LVDT which is an instrument for the measurement of displacement, works on the principal of ……..

1. Linear inductance
2. Non – linear inductance
3. **Mutual inductance**
4. Linear capacitance

Q93. The instruments used for the measurement of pressure is / are …….

1. Bellows
2. Diaphragms
3. Fiber optic pressure sensors

4. **All of these**
5. None of the above

Q94. A capacitive pressure sensor has a typical measurement uncertainty of …….

1. One
2. Two
3. Three
4. **Four**

Q95. If an instrument transformer is used to extend the ranges of AC instrument, then its reading will depend on ……..

1. R
2. L
3. C
4. All of these
5. **None of these**

Q96. The potential transformers are used to measure large voltage using ……..

1. High range voltmeter
2. Low range voltmeter
3. High range ammeter
4. **Low range ammeter**

Q97. Vibration galvanometer are generally used ……

1. For measuring electric charges
2. **As null – point detectors in ac bridges**
3. As null – point detectors in dc bridges
4. For measuring power

Q98. The two wattmeters used for the measurement of power input read 50 kW each. What will be the readings of the two wattmeters if the power factor is changed to 0.8 leading keeping the total input power same?

1. 28.35W, 71.65W
2. 31.25W, 73.71W
3. 33.33W, 73.33W
4. 38.35W, 75.5W

Ans. 1

Q99. For the measurement of unknown inductance in terms of known capacitance, the suitable ac bridges are ……..

1. Maxwell and Schering bridge

2. Maxwell and Wien's bridge
3. **Maxwell and Hay's bridge**
4. Hay's and Wien's bridge

Q100. The Wien's bridges is suitable for the measurement of frequency of the range of …….

1. Less than 100 Hz
2. **100 Hz to 100 kHz**
3. 1 kHz to 100 MHz
4. More than 100 MHz

Q101. For the measurement of low resistances, Kelvin's double bridge has high accuracy because:

1. **It has two set of ratio arms which eliminates effect of resistance of connecting lead**
2. It has a null indicating galvanometer
3. It has two null indicator
4. It has four sets of ratio arms which eliminates the effect of resistance of connecting lead

Q102. Swamping resistance is a resistance made up of ……

1. **Alloy of manganin and copper**
2. Alloy of aluminium and copper
3. Alloy of nickel and cobalt
4. Alloy of manganin and aluminium

Q103. In a moving coil of a meter swamping resistance is added to …….

1. Reduce the frequency error
2. **Reduce the temperature error**
3. Reduce the power consumption
4. All of these

Q104. The frequency errors in induction instruments can be compensated by the use of ……..

1. Non inductive shunt in both ammeters and voltmeters
2. **Non inductive shunt in case of ammeters and are generally self compensated in case of voltmeters**
3. Self compensated in case of both ammeters and voltmeters
4. Self compensated in case of ammeters and non inductive shunt in case of voltmeters

Q105. For reducing the errors in an induction instrument the alternating current to be measured has ……

1. **Same frequency with which the instrument was calibrated**
2. High frequency compared with which the instrument was calibrated
3. Low frequency compared with which the instrument was calibrated
4. None of these

Q106. In induction voltmeter, split phase windings are obtained by connecting a …….

1. High resistances in series with windings of both the magnets
2. **High resistance in series with the winding of one magnet and an inductive coil in series with the windings of other magnet**
3. An inductive coil in series with the winding of one magnet and a capacitance in series with the windings of other magnet
4. Inductive coils in series with the winding of both the magnets

Q107. A cylindrical type with split phase winding induction ammeters employs ……..

1. A rotating disc
2. **A hollow aluminum drum**
3. A single flux producing winding
4. Either (1) or (2)

Q108. Induction type instruments are generally used as …….

1. Ammeter
2. Voltmeter
3. Wattmeter
4. **All of these**
5. None of these

Q109. Damping torque is the torque which acts on ……..

1. Stationary system of the instrument
2. Moving system of the instrument only when it is stationary
3. **Moving system of the instrument only when it is moving**
4. Stationary system of the instrument only when the moving system is moving

Q110. The gravity controlled instruments has to be kept ……

1. **Vertical**
2. Horizontal
3. Inclined at 45 degree
4. Inclined at 75 degree

Q111. Random errors in a measurement system are due to ……..

1. **Environmental changes**

2. Use of uncalibrated instrument
3. Poor cabling practices
4. Unpredictable effects

Q112. Calibration of instrument is an important consideration in measurement system. The errors due to instruments being out of calibration can be rectified by ……..

1. **Increasing the frequency of recalibration**
2. Increasing the temperature coefficient
3. Increasing the susceptibility of measuring instrument
4. Decreasing the frequency of recalibration

Q112. The undesirable characteristics of a measuring system is …….

1. Drift
2. Dead zone
3. Non linearity
4. **All of these**

Ans. 4

Q113. If the instrument is used in wrong manner while application, then it will results in …….

1. Systematic error
2. **Instrument error**
3. Random error
4. Environmental error

Q114. Suitable method for the reduction of systematic errors is / are ……

1. Instrument must be designed carefully
2. By introducing an equal and opposite environmental input for compensating the effect of environmental input in a measurement system
3. By adding high gain feedback to measurement system
4. **All of these**

Q115. The systematic errors of an instrument can be reduced by making …….

1. **The sensitivity of instrument to environmental input as low as possible**
2. The sensitivity of instrument to environmental input as high as possible
3. Systematic errors does not depend on the sensitivity of instrument
4. None of these

Q116. The magnitude of environment – induced variation from the specified calibration condition is quantified by ……

1. Sensitivity drift
2. Zero drift
3. Backlash
4. **Both (1) & (2)**

Q117. In a moving coil voltmeter, the input resistance of the meter can be increased by …….

1. Increasing the number of turns in the coil
2. Decreasing the number of turns in the coil
3. Using the same number of coil turns made up of high resistance material
4. **Both (1) & (3)**

Q118. Depending on whether the display is a numeric or alphanumeric, the segmental display is ……

1. 7 segmental
2. 14 segmental
3. 21 segmental
4. **Either (1) or (2)**
5. None of the above

Q119. The measurements which can be simplified by using X – Y recorders is / are ……..

1. Speed – torque characteristics of motors
2. Regulation curves of power supplies
3. Hysteresis curves
4. **All of these**

Q120. Potentiometer is used for the measurement of ……..

1. Linear displacement
2. Angular displacement
3. Non – linear displacement
4. **Only (1) and (2)**
5. All the above

Q121. The displacement measuring instruments is / are ……

1. Potentiometer
2. LVDT
3. RVDT
4. **All of these**

Q122. In fibre – optic level sensors, the amount of light loss depends on ……..

1. **The proportion of cable that is submerged in the liquid**

2. Amount of light which is reflected back
3. The proportion of cable that is not in the liquid
4. Amount of light which is not reflected back

Q123. Instrument which is capable of discriminating temperature differences of even 0.1 degree Celsius is ……..

1. Fibre – optic level sensors
2. Laser method
3. **Thermography**

Q124. Contact devices used for the measurement of level are ……

1. Increases
2. Decreases
3. **Remains same**
4. None of these

Q125. For the measurement of pressure the instruments used can be …….

1. Mechanical
2. Electro – mechanical
3. Electronic
4. **All of these**
5. None of these

Q126. If at one end, the two wires made of different metals are joined together then a voltage will get produced between the two wires due to difference of temp between the two ends of wires. This effect is observed in ……..

1. **Thermocouples**
2. Thermistors
3. RTD
4. Ultrasonics

Q127. Schering bridges are used for the measurement of ……..

1. Unknown capacitance
2. Dielectric loss
3. Power factor
4. **All of these**
5. None of these

Q128. The AC Bridge which is used for the measurement of frequency is ……

1. Schering bridge
2. **Wien bridge**
3. Hay's bridge
4. Anderson bridge

Q129. If the reading of the two wattmeters is equal and opposite while measuring power in a 3 phase induction motor then the power factor of the load will be ……..

1. Unity
2. **Zero**
3. 0.5 lagging
4. 0.5 leading

Q130. The transformer ratio of the transformer depends upon the ……

1. Exciting current
2. Secondary current
3. Power factor of secondary circuit
4. **All of these**

Q131. Primary current in a current transformer is determined by ……

1. **The load on the system**
2. The load on its own secondary
3. The load on its own primary
4. All of these

Q132. The bridge suitable for the measurement of capacitance is / are ……

1. Anderson's bridge
2. Hay's bridge
3. Owen's bridge
4. **None of these**